

COA

Module-3

Pipelining and hazards

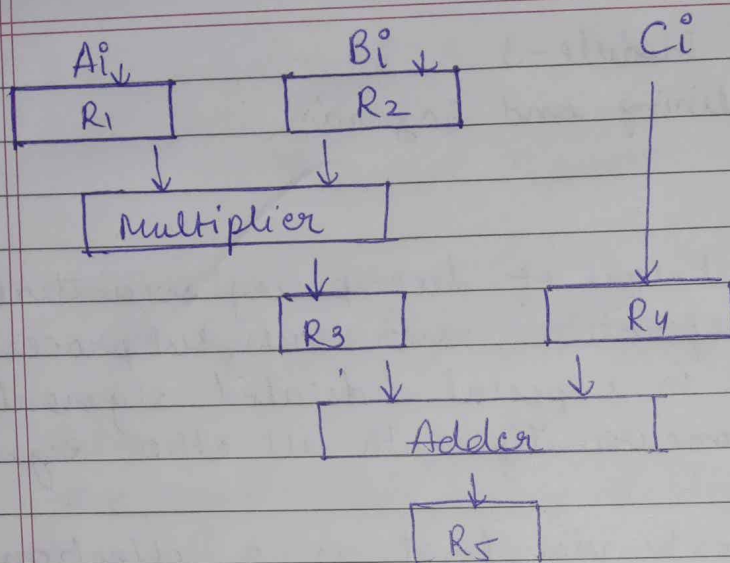
★ Pipelining:

- Pipelining is a technique of decomposing sequential process into suboperations, with each subprocess being executed in a special dedicated segment that operates concurrently with all other segments.
- A pipeline can be visualised as a collection of processing segments through which binary info. flows.
- Each segment performs partial processing dictated by the way the task is partitioned.
- The result obtained from the computation in each segment is transferred to the next segment in the pipeline.
- The final result is obtained after the data have passed through all segments.

$$A_i * B_i + C_i$$

where $i = 1, 2, 3, 4, \dots, n=7$.

- | | | |
|---|--|----------------------|
| ① | $R_1 \leftarrow A_i, R_2 \leftarrow B_i$ | 1/p A_i & B_i |
| ② | $R_3 \leftarrow R_1 * R_2, R_4 \leftarrow C_i$ | * & input C_i |
| ③ | $R_5 \leftarrow R_3 + R_4$ | Add C_i to product |



• Characteristic of Pipelining :

- 1) Instⁿ Overlap - Pipelining allows multiple instⁿ to be in diff. stages of execution simultaneously.
- 2) Increased throughput - By executing instⁿ in parallel, pipelining significantly ↑ the no. of instⁿ that can be complete per unit of time improving overall syst. performance.
- 3) Pipelining stages - The execution of instⁿ is divided into multiple stages, such as instⁿ fetch, decode, registers read, execute, memory access, & write back.

★ Types of Pipelining :

(i) Arithmetic Pipelining :-

It is specifically used for complex arithmetic opⁿs like floating point calculation.

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- It divides a complex opⁿ into smaller, simpler stages that can be executed concurrently, optimizing performance.

- Ex- A pipeline might be used to breakdown a *plication or float pt. opⁿ into stages like operand loading, *, & result writing.

(ii) Instruction Pipelining :-

- It breaks down the execution of instⁿ into a series of independent stages (eg- fetch, decode, execute).
- Multiple instⁿ can be in diff. stages of execution simultaneously, ↑ing throughput.
- Ex- A 5-stage instⁿ pipeline might involve fetching an instⁿ from memory, decoding, it accessing register, executing the opⁿ, then writing the result back to a register.
- * **Throughput** : Throughput refers to the amount of work a system can complete in a given time.
- * **Speed up** : Speedup measure the improvement in performance when using diff. architecture

$$S = \frac{n t_n}{(K + n - 1) t_p}$$

★ Pipeline Hazards :

- Pipeline Hazards is the situation that disrupts the sequential execution of instⁿ in a pipelined processor, potentially slowing down the pipeline.
- They are caused by dependencies b/w instⁿ that prevent the next instⁿ from executing in the subsequent clock cycle.

• Types of Pipeline Hazards :

(i) Structural Hazards:-

- It occurs when multiple instⁿ in the pipeline requires the same hardware resource, preventing parallel execution.
- For ex- If two instⁿs both need the same memory but at the same time, a structural hazard is created.

(ii) Data Hazards:-

- Arises when an instⁿ's execution depends on the result of a previous instⁿ that is still being processed in the pipeline.
- This can happen when one instⁿ needs the result of another instⁿ before the result is available, causing the pipeline to stall.

(iii) Control Hazards :-

Result from conditional branch instruction.

The branch target address might not be available in time, leading to pipeline stall. as the processor



waits for the target address to be determined.

• Resolving Pipeline Hazards:

- 1) Stalling - A simple approach is to pause the pipeline (insert no opⁿ (NOP) Instⁿ) until hazard is resolved
- 2) Forwarding - Passing the result of one instⁿ directly to another instⁿ that requires it, by passing the normal memory access path

* Parallel Processor:

- It refers to the execution of multiple instⁿ or tasks concurrently, rather than sequentially.
- It is a term used to denote a large class of techniques that are used to provide simultaneous data-processing tasks for the purpose of ↑ing the computational speed of comp system. Instead of processing each instⁿ sequentially as in conventional computer, a pp system is able to perform concurrent data processing to achieve faster execution time.
- For ex - while an instⁿ is being executed in the ALU, the next inst can be read from memory. The system may have two or more ALUs & be able to execute two or more processors operating concurrently.
- The main purpose of pp is to speed up the comp. processing capability and ↑es its throughput, that



is, the amount of processing that can be accomplished during a given interval of time.

- Key Aspects include simultaneous execution, ↑ speed, scalability.

• Types of Parallel Processing :

(i) Data Level -

- It divides a large dataset into small chunks & processes each chunk simultaneously using multiple processing unit.
- Widely used in big data processing, scientific simulation & machine learning.

(ii) Task Level -

- It divides a program into independent tasks that can be executed concurrently, each handling a specific part of the overall work load.
- It is used in multithreading application, real-time simulation.

(iii) Instⁿ Level :-

- It involve executing multiple instⁿ within a single clock cycle, improving the throughput of a single processor.

* **Concurrent Memory Access :** It refers to multiple threads or processes accessing same memory location at the same time.

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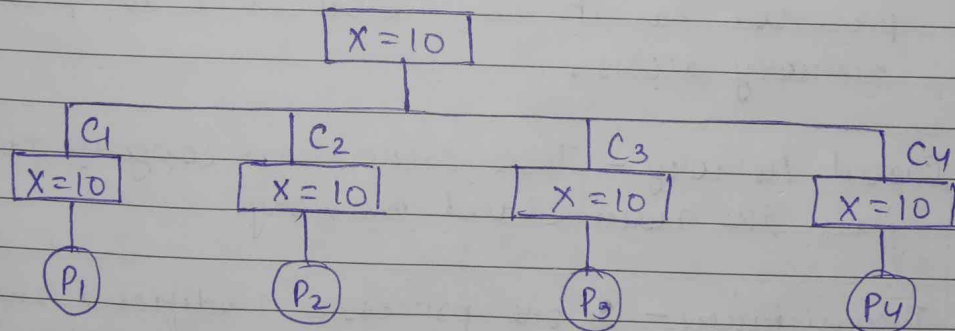
This can lead to corruption of data or race condition.

OR

It refers to the ability of multiple thread or processes to access and potentially modify the same memory location at the same location.

* Cache Coherence :

- Cache coherence ensures data consistency b/w system memory & multiple caches in a multi-core or multiprocessor system.
- When multiple processor or core access the same data, cache coherence protocols maintain a consistent view of the data across all cache, preventing data corruption & ensuring program correctness.



- 1) CMSI
 - 2) MOSI
 - 3) MESI
 - 4) MOESI
- C = Cache, E = Exclusive, M = Modify, S = Shared, I = Invalid

Questions.

(i) Explain write through & write policy in cache memory.

In cache memory, the write policy dictates how data is updated in both the cache & main memory.

- Write through ensure immediate update to both.
- Write back updates the cache first and may update main memory later.

2) Explain the cache coherence problems with solution in parallel processor.

The Problem:

- (i) Multiple Copies - In multiprocessor system, each processor has its own local cache to speed up memory access.
- (ii) Shared Memory - These cache stores copy of data from the main shared memory.
- (iii) Inconsistency - If one processor modifies a shared memory location, the changes might not be immediately reflected in other cache copies, leading to inconsistencies.

The Solution:

- (i) Cache Coherence Protocols - These protocols manage data consistency by ensuring that all caches

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have a consistent view of shared memory.

(ii) Shared Cache - Instead of each processor having its own cache, a shared cache can be used, simplifying the coherence problem.

(iii) Update - Instead of Invalidation, the cache line in other cache can be update directly with the new value, maintaining consistency.

3) Write the advantages of write through policy.

1) Data Consistency

2) Safety

4) Draw a space time diagram for 4-segment pipeline shown 6 tasks.

Time required to complete all operation.

$$4 + (6-1) = 4 + 5 = 9 \text{ clock cycles.}$$

	1	2	3	4	5	6	7	8	9
Segment 1	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆			
Segment 2		T ₁	T ₂	T ₃	T ₄	T ₅	T ₆		
Segment 3			T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	
Segment 4				T ₁	T ₂	T ₃	T ₄	T ₅	T ₆

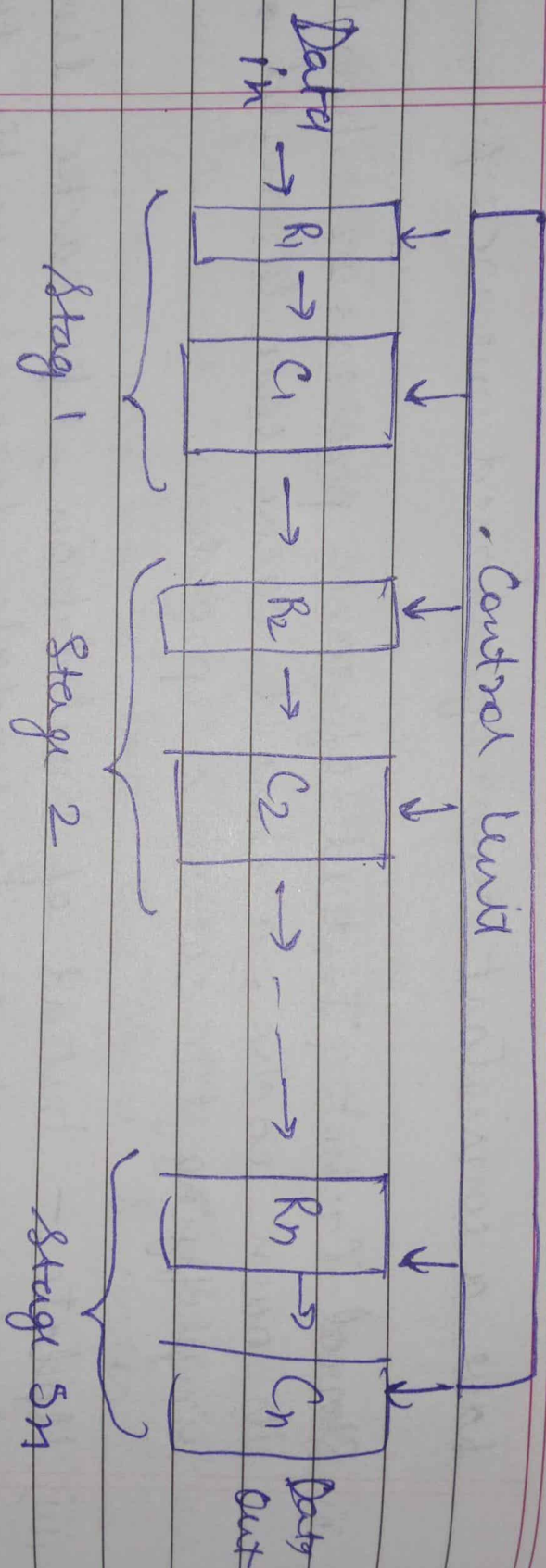
5) Draw the diagram of pipelining?

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Formula of throughput :

$$TH = \frac{1}{T}$$